

Review on Torque Ripple Reduction Techniques of BLDC Motor

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Abstract:

This paper reviews the torque ripple reduction techniques of Brushless DC motor (BLDC). Torque ripple arises in BLDC motor due to the attraction of rotor's permanent magnets and stator's ferro magnet teeth. The magnetic attraction varies depending on the strength of the magnetic field and this dissimilarity leads to the production of irregular torque called torque ripples. This torque ripple may cause poor quality of speed-torque characteristics, noise, vibrations and produce stern faults in sensor less motor drives. This paper analyses various techniques for the reduction of torque ripples. The analysis shows that the torque ripple in the BLDC motor can be minimized during commutation time by increasing the input voltage four times greater than the back emf. Also, this paper reviews different methodologies to boost the input voltages during the commutation period.

Keywords:

BLDC motor, Torque ripple, Cuk converter, Charged capacitor, DVM scheme

I.INTRODUCTION

In the early 19th century, PM excitation was not implemented in electrical machines because of the poor quality of permanent magnet (PM) materials. Alnico's invention rejuvenated the PM excitation system but it was restricted to small and fractional horse power dc machines [1]. The squirrel cage induction motor became popular in industrial applications due to the advances in power electronics. PM brushless DC (PMBLDC) motor was developed due to the poor efficiency and power factor of the squirrel cage induction motor than the synchronous motor and speed, noise and EMI problems of the synchronous motor.

BLDC motors have the advantages of high efficiency, simple construction, higher dynamic performance, higher speed versus torque characteristics, longer operating life, lower noise and higher speed range [17].

PMBLDC motors can be used in applications with constant load such as fans, blowers, pumps etc. BLDC motor also gives high precision and good dynamic response in the varying load applications. The few examples of varying loads

are home appliances such as washers, dryers, automotive applications such as engine control, electric vehicle control and fuel pump control, centrifugal aerospace applications, pumps, robotic arm control and gyroscope control [17]. Recent advances in control technology have also made use of the BLDC motor in the field of machine tool position control and industrial robots.

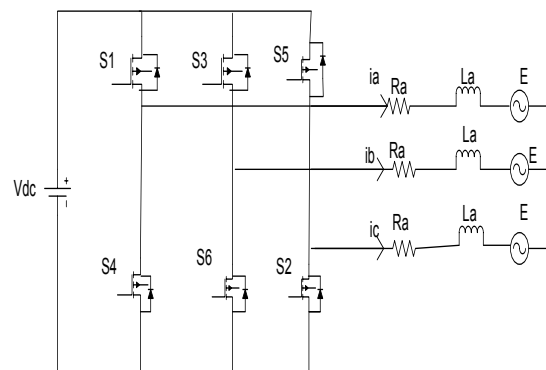


Fig1.Equivalent circuit of BLDC motor

Fig.1 illustrates the equivalent circuit of BLDC motor. Since the permanent magnet BLDC motor is brushless, the hall sensors are positioned on the stator winding, to obtain the rotor position [5]. The BLDC motors generate trapezoidal back-emf and it is applied with rectangular stator currents [6]. However, pure trapezoidal wave is not obtained due to the attraction of permanent magnets of rotor and the ferro magnet teeth of stator. Depending on magnetic field strength the magnetic attraction varies. This magnetic attraction variation produces ripple torque in the BLDC motor. This torque ripple can cause poor quality of speed-torque characteristics, noise, vibration and produce stern faults in sensor less motor drives [5]. BLDC motor has two incoming and outgoing phases and one floating phase. In

general, the growing rate of incoming phase current and decay of outgoing phase current are not equal which raises the torque ripple.

Numerous topologies have been proposed to reduce the ripple of torque in the BLDC engine. The Z-Source inverter buck-boost function provides a good condition for the reduction of BLDCM's commutation torque ripple using the two-phase modulation mode [2]. Depending on the study of torque variance rates [9], a non-ideal back emf and current commutation is debated. To increase the dc-link voltage, the quasi-Z source is used by changing the shoot-through ratio of the switch [16]. A modified smooth torque ripple reduction is constructed based on the back-EMF variation [7]. A novel commutation sequence is proposed for the reduction of torque ripple in which the incoming phase current can build up slowly and reaches the set value [10]. A new topology on linearization of the input-output feedback is suggested to reduce the ripple torque [13].

II. COMMUTATION TORQUE RIPPLE ANALYSIS

The analysis of commutation torque ripple can be accomplished by considering phase A as incoming and phase B as outgoing.

The expression for developed electromagnetic torque is given by,

$$T_d = \frac{e_a i_a + e_b i_b + e_c i_c}{\omega} \quad (1)$$

$$i_a = -i_c; i_b = 0$$

$$i_a = -i_c = I \quad (2)$$

Where

e_a, e_b, e_c = Back emf (V)

i_a, i_b, i_c = Stator current per phase(A)

ω = Rotor speed (rad/sec)

T_d = developed electromagnetic torque (Nm)

T_L = Load torque (Nm)

A. Before commutation

Assume constant back emf, then T_d is given by,

$$T_d = \frac{E i_a + 0 - E(-i_a)}{\omega} = \frac{2E i_a}{\omega} = \frac{-2E i_c}{\omega} \quad (3)$$

Under steady state condition the current $i_a = -i_c = I$ = Constant for the constant T_L and constant speed.

B. After Commutation

$$T_d = \frac{2E i_b}{\omega} = \frac{-2E i_c}{\omega} \quad (4)$$

$$i_b = -i_c = I = \text{Constant}$$

C. During Commutation

During commutation period the switch S1 is OFF, S3 is ON, diode d4 starts conducting and switch S2 continues to conduct. Applying KVL to the circuit,

$$R_a i_a + L_a \frac{di_a}{dt} + E + V_n = 0 \quad (5)$$

$$R_a i_b + L_a \frac{di_b}{dt} + E + V_n = V_{dc} \quad (6)$$

$$R_a i_c + L_a \frac{di_c}{dt} - E + V_n = 0 \quad (7)$$

$$T_d = \frac{2E(i_a + i_b)}{\omega} = \frac{-2E i_c}{\omega} \quad (8)$$

Hence the current i_c will remain constant for the condition of,

$$\frac{di_a}{dt} = -\frac{di_b}{dt} \quad (9)$$

D. Condition for minimum torque ripple

$$L_a \frac{di_c}{dt} = E - V_n - R_a i_c \quad (10)$$

Neglecting the resistance drops and adding (5),(6) and (7), the non-commutating phase current equation is given by,

$$3L_a \frac{di_c}{dt} + E + 3V_n = V_{dc}$$

$$3L_a \frac{di_c}{dt} = V_{dc} - E - 3V_n$$

$$\frac{di_c}{dt} = \frac{V_{dc} - E - 3V_n}{3L_a} \quad (11)$$

For steady state,

$$\begin{aligned} \frac{di_a}{dt} &= -\frac{di_b}{dt} \\ \frac{-V_n - E}{L_a} &= -\left(\frac{-V_n - E}{L_a}\right) \\ -V_n - E - V_n - E &= 0 \\ -2V_n - 2E &= 0 \\ V_n &= E \end{aligned} \quad (12)$$

Substituting in (11),

$$\begin{aligned} \frac{di_c}{dt} &= \frac{V_{dc} - 4E}{3L_a} \\ \frac{di_c}{dt} &= 0 \end{aligned}$$

$$V_{dc} = 4E$$

$$(13)$$

$$\begin{aligned} \Delta T_d &= \frac{-2E i_c}{\omega} \\ &= \frac{2E}{\omega} \left[\frac{-V_{dc} + 4E}{3L_a} \right] \end{aligned} \quad (14)$$

E. Expression for the duty cycle during conduction period

Before commutation, the voltage equations are given by,

$$R_a i_a + L_a \frac{di_a}{dt} + E + V_n = V_{dc \text{ nom}} \delta_{\text{nom}} \quad (15)$$

$$R_a i_c + L_a \frac{di_c}{dt} - E + V_n = 0 \quad (16)$$

From (15)-(16)

$$R_a i_a + L_a \frac{di_a}{dt} + E + V_n - R_a i_c - L_a \frac{di_c}{dt} + E - V_n = V_{dc} \delta_{nom} \quad (17)$$

$$i_a = -i_c$$

$$V_{dc} \delta_{nom} = -2R_a i_c + 2E$$

$$\delta_{nom} = \frac{-2R_a i_c + 2E}{V_{dc}} \quad (18)$$

Equation (18) gives the expression of duty cycle for the conduction period δ_{nom} .

From (13), it is proved that the dc voltage should be four times higher than the back-emf to reduce the torque ripple during commutation time. This paper focuses on various techniques for the reduction of torque ripple viz., Cuk converter method, charged capacitor and modulator voltage scheme. Using these methods, the dc voltage value is increased 4 times more than the back emf.

III. TORQUE RIPPLE REDUCTION TOPOLOGIES:

A. CUK Converter

The Cuk converter is a DC-DC converter with a boost converter and a buck converter followed [18]. Hence, the input voltage can be boosted or minimized depending on the duty cycle value. This boosting cuk converter feature can be used to lower the BLDC motor torque ripple.

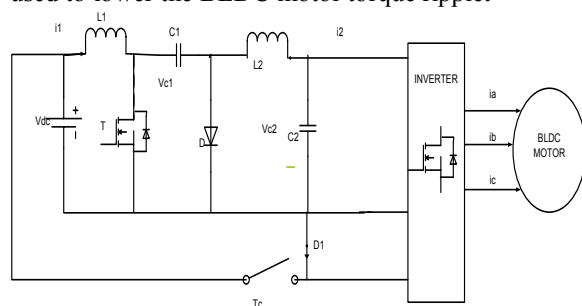


Fig.2 Mode selection circuit of Cuk converter in mode 1 during normal operation

The circuit operation can be divided into two modes [18]. As shown in Fig.2, during the mode 1 operation, the switch T is OFF and the voltage across C1 forward biases diode D. Under this condition, the cuk converter supplies energy to the motor through diode D1. The inductors L1, L2 stores energy and it can be expressed as,

$$L1 \frac{di_1}{dt} = V_{dc}$$

$$L2 \frac{di_2}{dt} = V_{c1} - V_{c2} \quad (19)$$

Where

i_1 and i_2 = currents through L1 and L2
 V_{c1} and V_{c2} = voltages across C1 and C2

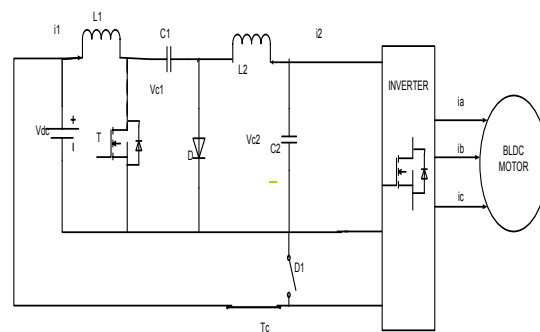


Fig.3. Mode selection circuit of cuk converter during commutation period

Fig.3 illustrates, during the mode 2 operation, the switch T is ON and the voltage across C1 reverse biases diode D. Under this condition, the cuk converter supplies energy to the motor through switch Tc and diode D1. The inductors L1, L2 supplies energy and it can be expressed as,

$$L1 \frac{di_1}{dt} = V_{dc} - V_{c1}$$

$$L2 \frac{di_2}{dt} = -V_{c2} \quad (20)$$

Where,

V_{c1} and V_{c2} = average voltage of C1 and C2

d_c = duty cycle.

$$V_{c1} (1 - d_c) = V_{dc} \quad (21)$$

$$V_{c2} = d_c V_{c1} \quad (22)$$

From (20), during mode 2 operation the output voltage is increased than the supply voltage.

The experimental set of this proposed system is experimentally implemented on a 70W, 24V, 4A, 3000rpm and load torque of 0.23N-m BLDC motor [4].

B. Charged Capacitor Method

This topology consists of a capacitor C, a diode D, a resistance R and a mode selection switches S_1 and S_2 . The capacitor C is charged, during the normal conduction period. Fig.4 illustrates the circuit diagram of charging capacitor scheme.

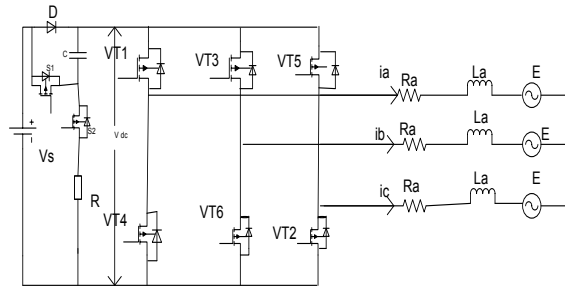


Fig.4. Charging capacitor scheme

The capacitor C is charged through diode D during normal conduction period. Under this condition, the switch S1 is OFF and S2 is ON. Once the capacitor C is fully charged, then S2 is OFF and S1 is ON. Now commutation starts and diode D is reverse biased. Under this condition the supply voltage is used to increase the dc link voltage to reduce the torque ripple.

To reduce the torque ripple, the voltage V_{dc} has to satisfy the following condition,

$$V_{dc} = V_s + U_e \geq 4E \quad (23)$$

Also the value of capacitor voltage during the commutation is,

$$U_e \geq 4E - V_s \quad (24)$$

This topology is experimentally tested with a rating of 3.7kW, 300V, 1500rpm, 24 Nm BLDC motor [3]. When the V_{dc} is equal to 300V, the non-commutated phase current relative current fluctuation is 23.4% and the torque ripple is 20.4%. The advantage of this proposed scheme is the requirement of chopper circuit is avoided.

C. Voltage Modulator Scheme

As shown in Fig.5, the proposed topology consists C1 and C2, L, S1 and S2 and diode D[8].

During the normal conduction period, the switch S1 is open and S2 is closed. Both the capacitors are connected across the supply and hence supplies DC voltage to the inverter.

During commutation period, S1 is closed and S2 is open. Under this condition both capacitors are in series and supplies double the value of Dc voltage to the inverter [8]. Thus the supply voltage is increased to reduce the torque ripple of the BLDC motor.

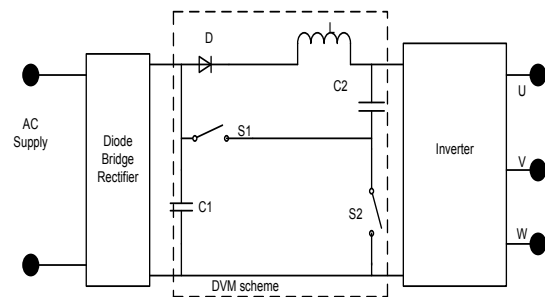


Fig.5. DVM scheme

This proposed scheme is experimentally tested for 1hp, 300V, 2500rpm and load torque of 2Nm PMBLDC motor. The DVM circuit includes 4700μF capacitors and 10mH inductor [8]. This proposed DVM scheme is used to reduce the torque ripple during high speed operation of BLDC motor. Also pulse width modulation control is used for low speed operation.

Table 1 gives the details of performance parameters of different torque ripple reduction topologies.

Parameters		Cuk Converter [4]	Charged Capacitor [3]	DVM Scheme [8]	Auxiliary step up circuit [12]	PWM [11]	Three level NPC[6]
PMBLDC Motor	Rated Voltage (V)	200	300	300	200	200	200
	Rated power(kW)	70W	3.7	1hp	3.2	70W	1hp
	Rated Speed(rpm)	3000	1500	2500	1500	1800	3000
	Rated torque Nm	0.23	24	2	20	20	0.825
	Rated current (A)	4	3	3	4	3	5
	Phase inductance (mH)	0.61	1.88	21.5	1.234	0.6	
Torque ripple reduction circuit	Inductance	L1=L2=330μH	-	10mH	-	-	-
	Capacitance(μF)	C1=1100μF C2=2200μF	120μF	4700 μF	60	-	-
	Resistors(KΩ)	-	-	-	-	Voltage divider R ₁ =6.04, R ₂ =2	-
	Switches (S +D)	8+2	8+1	8+1	8+1	6	12+6
	Switching frequency(kHz)	20	20	20	20	20	80

Table1. Performance parameters of different torque reduction topologies

IV. CONCLUSION

This paper discusses about the analysis of commutation torque ripple of BLDC motor. The torque ripple in the BLDC motor has been proven to be minimized during commutation time by increasing the input voltage four times greater than the back emf. Also, this paper reviews various methods of torque ripple reduction in PM BLDC motor viz., Cuk converter, charged capacitor and voltage modulation schemes. The cuk converter operates in boost mode during the commutation period to increase the input voltage of the inverter while the capacitor voltage is added to the input voltage in the charged condenser method. The DVM scheme is used to reduce the torque ripple during high speed operation of BLDC motor and pulse width modulation control is used for low speed operation. Also, performance parameters of different torque ripple reduction topologies were tabulated.

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